

i When to Use

To test for a relationship between two categorical/qualitative variables when at least one variable has more than 2 categories.

H Research Hypothesis: The researcher hypothesized that there is a pattern of relationship between the height gap category and the age difference category in Superman portrayals. Specifically, portrayals with minimal height gaps would tend to also have minimal age differences, while portrayals with big height gaps would tend to have big age differences.

H Null Hypothesis: There is no pattern of relationship between height gap category (Minimal, Average, Big) and age difference category (Minimal, Average, Big) in the population represented by this sample.

How To

Analyze → Descriptive Statistics → Crosstabs

- Highlight the variable you want to define the rows (be sure it is qualitative/categorical) and click arrow
- Highlight the variable you want to define the columns (be sure it is qualitative/categorical) and click arrow
- “Statistics” — check that you want a “Chi-square analysis”

SPSS Syntax

```
CROSSTABS
  /TABLES=height_gap BY age_grp
  /STATISTICS=CHISQ
  /CELLS=COUNT ROW COLUMN TOTAL.
```

- ① “row variable” BY “column variable”
- ② Get Chi-square significance test
- ③ Get various row, column and/or total cell percentages (optional)



SPSS Output

Chi-Square Tests

	Value	df	Asymptotic Significance (2- sided)	Exact Sig. (2- sided)
Pearson Chi-Square	39.058	4	<.001	
Fisher-Freeman-Halton Exact Test				<.001
N of Valid Cases	47			

6 cells (66.7%) have expected count less than 5. The minimum expected count is 1.79.
Because expected counts are sparse, use the Fisher-Freeman-Halton exact two-sided p-value for inference.

height_gap * age_grp Crosstabulation

height_gap		Minimal	Average	Big	Total
Minimal	Count	5	0	7	12
	Expected Count	3.1	4.3	4.6	12.0
Average	Count	7	0	0	7
	Expected Count	1.8	2.5	2.7	7.0
Big	Count	0	17	11	28
	Expected Count	7.1	10.1	10.7	28.0
Total	Count	12	17	18	47
	Expected Count	12.0	17.0	18.0	47.0

Assuming the null hypothesis is true, a p-value is the probability of obtaining a table at least as inconsistent with independence as the observed table. It is not the probability that the result is a Type I error.

SPSS may display **.000** when a very small p-value rounds to three decimal places. Report that value as **p < .001**, not p = .000.

In this example, 6 of 9 cells (66.7%) have expected counts below 5, so the Pearson chi-square approximation is not used for inference. The reported result uses the Fisher–Freeman–Halton exact test, the extension of Fisher’s exact test to an r×c table.

Follow-Up Analyses

Note

When a k×k chi-square test is significant, follow-up 2×2 chi-square tests are conducted for each pairwise comparison. These follow-ups help determine which specific combinations of categories are driving the overall pattern. Note that these follow-ups have less power than the original test, so it is possible to find a significant overall pattern but have none of the pairwise comparisons be significant.

Having found an overall effect, we need to perform follow-up analyses to test if the pattern of data matches the research hypothesis.

To do this we need to look at pairwise comparisons, specifically, looking at each 2×2 table.

Minimal vs Big height gap × Minimal vs Big age difference

height_gap * age_grp Crosstabulation

height_gap		Minimal	Big	Total
Minimal	Count	5	7	12
	Expected Count	2.6	9.4	12.0
Big	Count	0	11	11
	Expected Count	2.4	8.6	11.0
Total	Count	5	18	23
	Expected Count	5.0	18.0	23.0

Reporting the Results

Sample APA Write-Up

For the sample of 47 Superman portrayal ratings, participants were classified by height gap category and age difference category. The Fisher–Freeman–Halton exact test indicated a significant relationship between height gap and age difference, $p < .001$. Follow-up 2x2 chi-square analyses were conducted to determine which pairwise comparisons contributed to the overall pattern. The Minimal vs Big comparison was significant, Fisher's exact test, $p = .037$.

Key Reporting Elements

When reporting k x k chi-square results, be sure to communicate:

- Sample size and how participants were classified
- The overall chi-square test result (chi2, df, p)
- The follow-up pairwise 2x2 analyses (if the overall test was significant)
- The direction of each significant pairwise relationship
- Whether or not the pattern of results supports the research hypothesis